# Yiwen Chen

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## Education

**University of Chicago** Sep. 2026 – Present
*M.S. in Data Science* Chicago, IL

**University of Minnesota, Twin Cities** Sep. 2023 – May 2026
*B.S. in Statistics; Minor in Mathematics* Minneapolis, MN
GPA: 3.9/4.0

**Southwest University of Political Science and Law** Sep. 2021 – Jun. 2023
*Economic Statistics (transferred)* Chongqing, China

## Research Interests

Computational social science; educational inequality and opportunity; family contexts and adolescent development; social determinants of health; statistical modeling; interpretable machine learning.

## Research Experience

**Technology, Teens, and Families Lab, University of Minnesota** Sep. 2024 – 2026
*Undergraduate Research Assistant, advised by Prof. Xiaoran Sun* Minneapolis, MN

- Studied how adolescents' family experiences relate to educational achievement using Add Health data, applying unsupervised learning to identify distinct family-experience profiles and link them to educational outcomes.
- Contributed across the research process: developing research questions and study design, constructing variables in R, handling missing data, clustering analysis, interpreting results, and drafting the manuscript.
- Conducted a literature review on overnight smartphone use, summarizing measurement approaches, statistical methods, and key empirical findings to inform the team's research plans.

**Oak Ridge National Laboratory** May 2025 – Aug. 2025
*Pathways to Computing Internship Program (PCIP), advised by Dr. Seung-Hwan Lim and Dr. Guojing Cong* Oak Ridge, TN

- Characterized cross-species toxicogenomic data through exploratory analysis of human and rat transcriptomic datasets (gene–gene correlations, fold-change concordance, descriptive statistics).
- Applied the Kolmogorov–Smirnov test to compare gene-expression distribution shifts across datasets, identifying genes driving major between-species differences as interpretable targets for downstream analysis.
- Evaluated the feasibility of integrating DrugMatrix and Open TG-GATEs by organizing raw microarray data into analysis-ready structures and comparing cross-platform expression distributions after unified RMA preprocessing.

**Summer Program for Undergraduate Research (SPUR), University of Utah** May 2024 – Aug. 2024
*Summer Undergraduate Researcher, advised by Prof. Andy Hong and Prof. Katherine E. Isaacs* Salt Lake City, UT

- Examined how environmental factors relate to outdoor fall injuries among older adults by linking national EMS (NEMSIS) injury records with ZIP-code-level environmental data and fitting a zero-inflated negative binomial model.
- Mapped the geographic distribution of fall incidents and used R and Tableau to visualize temporal trends, spatial patterns, and demographic differences in fall risk; results were published in *RANGE: Undergraduate Research Journal*.

## Publications & Presentations

*Journal Article*

[1] **Chen, Y.**, Hong, A., & Isaacs, K. (2024). Environmental determinants of outdoor fall injuries among older adults: A national study. *RANGE: Undergraduate Research Journal*. [Online article]
Also presented as a poster at the University of Utah SPUR Undergraduate Research Symposium, 2024.

*Conference Papers & Posters*

[2] **Chen, Y.**, Cong, G., Lim, S.-H., Chao, F., & Auerbach, S. S. (2026, March). *Statistical evaluation of DrugMatrix and Open TG-GATEs for construction of an integrated toxicogenomics resource* [Poster, Abstract No. 4385]. Society of Toxicology (SOT) Annual Meeting, San Diego, CA.

[3] Chen, S., **Chen, Y.**, Xu, T., Lee, C. T., & Sun, X. (2025, November). *From family context to achievement in diverse adolescents: Uncovering hidden family experience patterns with unsupervised machine learning* [Paper presentation]. National Council on Family Relations (NCFR) Annual Conference, Baltimore, MD. *Manuscript under review.*

[4] **Chen, Y.**, Cong, G., & Lim, S.-H. (2025, July). *Toxicogenomic variation in cross species and cross mode studies* [Poster]. Oak Ridge National Laboratory PCIP Poster Session, Oak Ridge, TN.

## Selected Projects

**Normalizing Flows for Gravitational-Wave Parameter Estimation** Spring 2025
*Collaborative course project, AST 8581: Big Data Analytics in Astrophysics, University of Minnesota* [GitHub]

- Explored likelihood-free Bayesian parameter estimation for gravitational-wave signals: time-frequency representations of LIGO Hanford and Livingston data were encoded by a similarity embedding network (VICReg objective) and used as context for a masked autoregressive normalizing flow.
- Contributed flow data preparation, training and testing components, and diagnostic visualizations of learned representations and parameter estimates.

**Cross-Database Integration of Rat Liver In Vivo Toxicogenomics Data** Spring 2026
*Statistics capstone, University of Minnesota*

- Evaluated systematic batch differences between DrugMatrix and Open TG-GATEs rat liver gene-expression data, using PCA, distributional analysis, and a control-anchor correction strategy to make the data more comparable without erasing biologically meaningful signals.

## Honors & Awards

- **Buehler Memorial Undergraduate Award** ($1,500), School of Statistics, University of Minnesota Sep. 2025
  Awarded annually to 3–5 Statistics undergraduates for excellence in statistical modeling and technical communication.
- **CLA Waller Scholarship** ($2,000), University of Minnesota May 2025
- **First Place, Undergraduate Category**, ASA Twin Cities Chapter Spring Poster Competition ($150) Apr. 2025
  For the poster "Environmental Determinants of Outdoor Fall Injuries among Older Adults: A National Study."
- **Undergraduate Research Opportunities Program (UROP) Award** ($900), University of Minnesota Nov. 2024
- **Dean's List**, University of Minnesota 2023 – 2025
- **Third Prize, Chongqing Region**, National College Student Statistical Modeling Competition, China May 2023
- **Merit Scholarship**, Southwest University of Political Science and Law Mar. 2022

## Outreach & Activities

**Volunteer Community Math and English Tutoring** Oct. 2021 – Apr. 2022

- Provided individualized mathematics and English tutoring to children in rural communities through weekly online sessions, adapting explanations and activities to each student's needs.

**Midwest Undergraduate Data Analytics Competition (MUDAC)** Apr. 2024 & Apr. 2025

- 2025: Conducted text-based sentiment analysis of call-record notes from Big Brothers Big Sisters of Minnesota using Python and NLTK to identify themes and emotional signals.
- 2024: Investigated causes of missing agricultural data across Minnesota counties and built geographic and temporal visualizations of statewide agricultural indicators in Tableau.

**Oak Ridge Computing Academy (ORCA): HPC System Administration Training, ORNL** Jul. 2024

- Week-long hands-on training at the Oak Ridge Leadership Computing Facility on cluster setup, virtualized HPC nodes, and networking.

## Skills

**Programming & analysis:** Python, R, Stata, SPSS · **Machine learning:** PyTorch, scikit-learn, pandas
**Visualization:** Tableau, ggplot2, matplotlib · **Tools:** Git, LaTeX, MongoDB, Linux